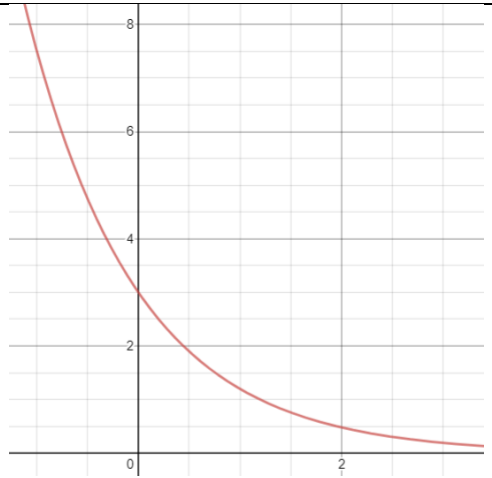
Question			Answer	Marks	AO	Guidance
3	(a)			M1 B1 A1	1.1 1.1 1.1	decreasing concave up curve in 1st and 2nd quadrants which does not cut the x-axis; mark intent decreasing curve with intercept (0,3); may be in one quadrant only smooth curve from (–0.5, a) through (2.5, b), where $4.5 \leq a \leq 5$ and $0 < b < 0.5$
				[3]		
3	(b)		$\log(3 \times 0.4^x) = \log(0.8)$ **oe** $x\log 0.4 = \log 0.8 - \log 3$ **oe** 1.44 **cao**	M1 M1 A1	3.1a 1.1 1.1	taking logarithms in any base 3rd law of logs used correctly if **M0M0** allow **SC1** for 1.44 unsupported
			Alternatively $0.4^x = \frac{0.8}{3}$ $x = \log_{0.4}\left(\frac{0.8}{3}\right)$ x = 1.44 **cao**	M1 M1 A1		 may see $x\log 0.4 = \log\left(\frac{0.8}{3}\right)$ **oe**
				[3]		